

New Tyres Inventory Analysis

Inventory Optimization and Product Performance Analysis

Georgina Casas

Google Data Analytics Professional Certificate

Portfolio Project

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Business Problem

Inventory management is critical for tire distributors.

Key challenges include:

- Excess inventory tying up capital
- Risk of stockouts for high-demand products
- Limited visibility into product performance

Goal:

Use inventory and sales data to support better purchasing decisions.



Dataset and Methodology

Dataset

- 18,000 inventory records
- 200 tire products
- 12 tire size categories
- 90-day analysis period

Tools

- Python
- Pandas
- Matplotlib
- Google Colab
- GitHub

KPIs Calculated

- Total Sales
- Average Inventory
- Inventory Turnover
- Inventory Value
- Product Classification

Key Findings

36,058

Units Sold

295/80R22.5

Highest Sales Volume

46.65

Average Inventory

275/65R18

Highest Inventory Investment

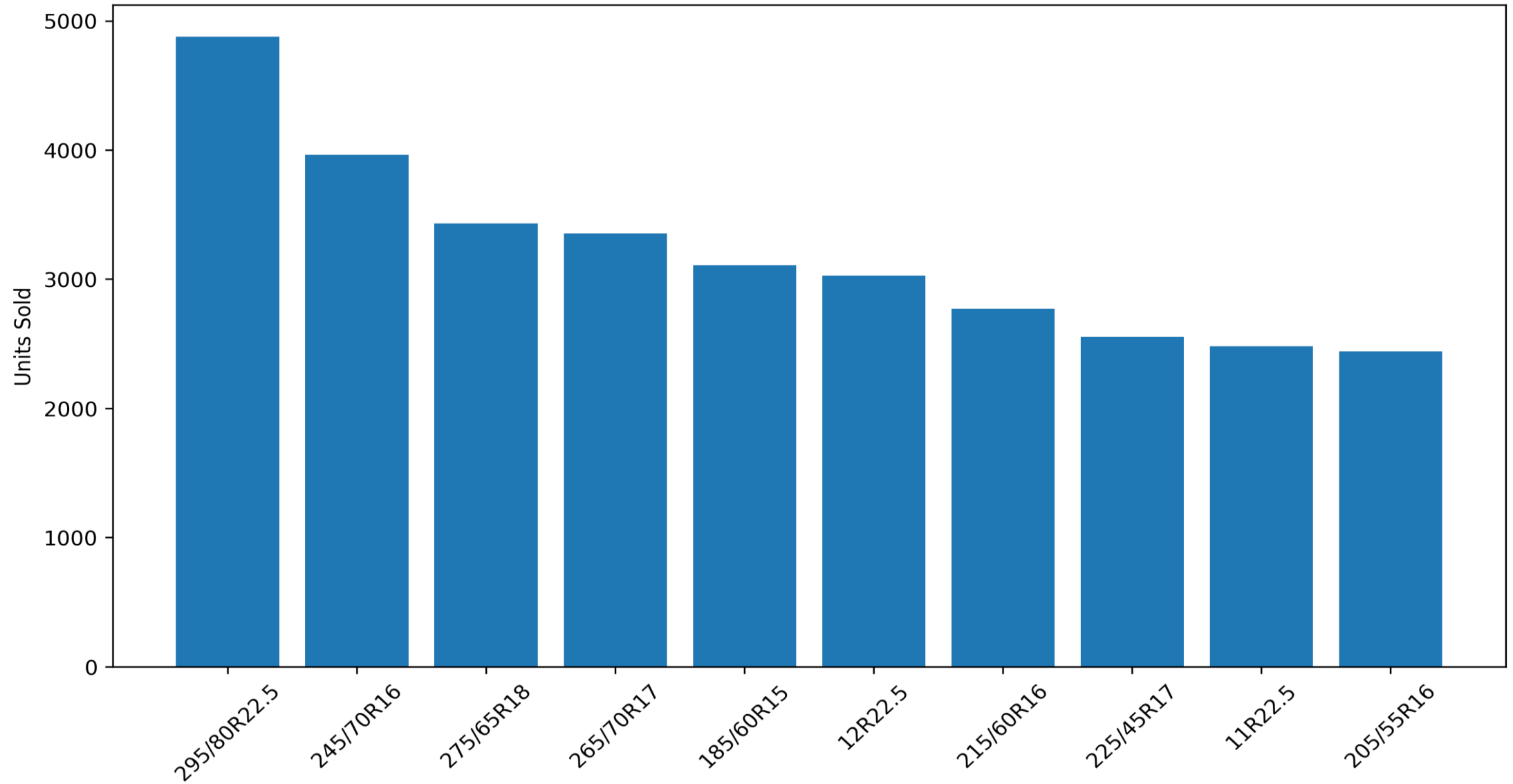
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Star Products

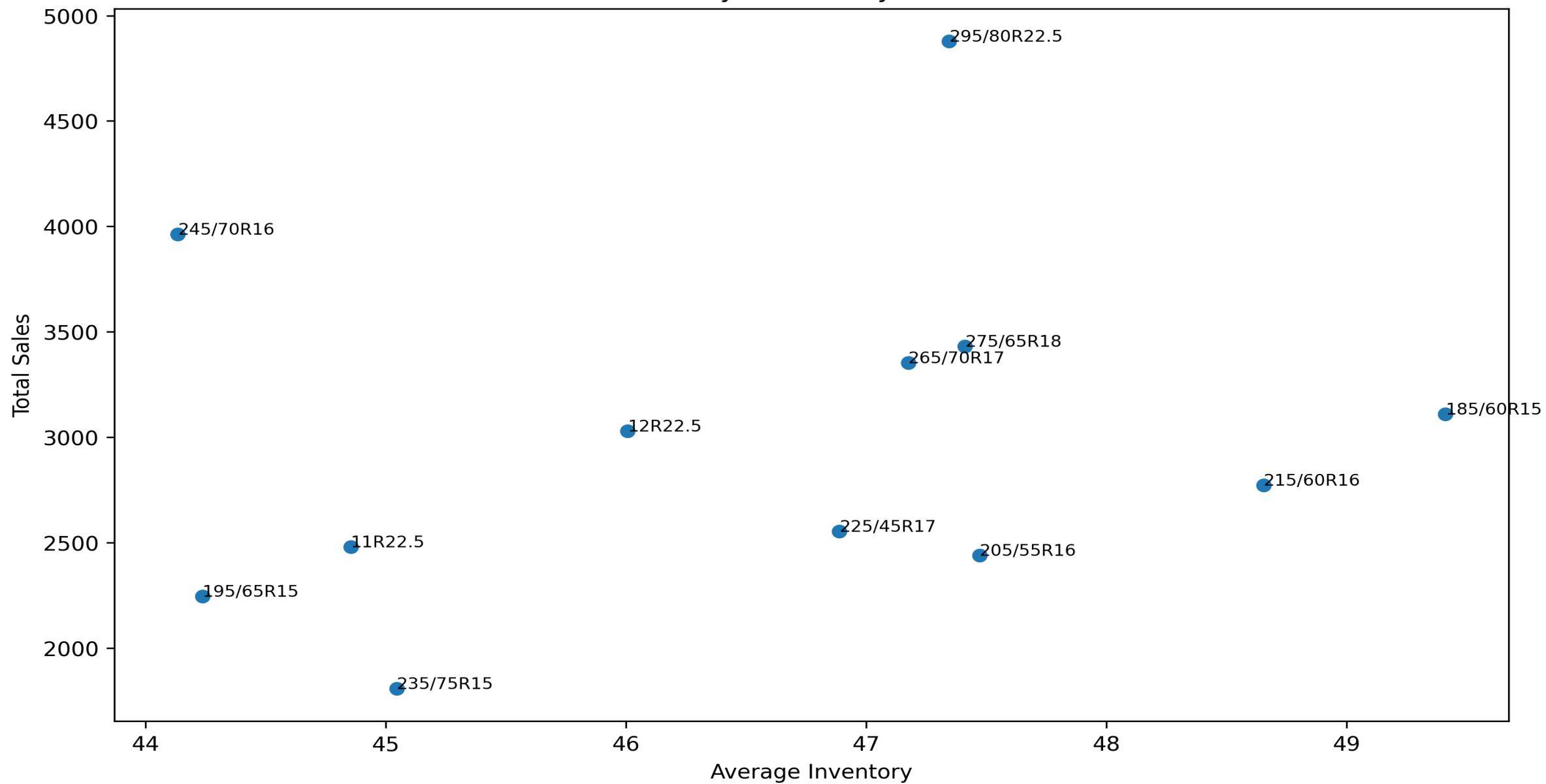
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Slow-Moving Products

Top 10 Tire Sizes by Sales



Inventory vs Sales by Tire Size



Recommendations

1. Prioritize replenishment of high-performing products.
2. Monitor inventory investment in high-value products.
3. Reduce purchasing frequency for slow-moving products.
4. Continue developing inventory analytics to support decision-making.

This analysis demonstrates how inventory data can be transformed into actionable business insights.



Thank You

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